Network Protocols Assignment

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#	Protocol	Layer	Port	TCP/UDP	Encryption	Usage
1	HTTP	Application	80	TCP	No	Web browsing
2	HTTPS	Application	443	TCP	Yes	Secure web browsing
3	FTP	Application	21	TCP	No	File transfer
4	FTPS	Application	990	TCP	Yes	Secure FTP
5	SFTP	Application	22	TCP	Yes	Secure file transfer
6	SSH	Application	22	TCP	Yes	Remote login
7	Telnet	Application	23	TCP	No	Remote access
8	SMTP	Application	25	TCP	No	Email sending
9	SMTPS	Application	465	TCP	Yes	Secure email sending
10	POP3	Application	110	TCP	No	Email retrieval

11	POP3S	Application	995	TCP	Yes	Secure email retrieval
12	IMAP	Application	143	TCP	No	Email management
13	IMAPS	Application	993	TCP	Yes	Secure email
14	DNS	Application	53	UDP/TCP	No	Domain resolution
15	DoH	Application	443	TCP	Yes	DNS over HTTPS
16	DHCP	Application	67/68	UDP	No	IP assignment
17	SNMP	Application	161	UDP	No	Network monitoring
18	SNMPv3	Application	161	UDP	Yes	Secure monitoring
19	NTP	Application	123	UDP	No	Time sync
20	RTP	Application	Dynamic	UDP	No	Streaming
21	RTCP	Application	Dynamic	UDP	No	Streaming control
22	RTSP	Application	554	TCP/UDP	No	Media streaming

23	SIP	Application	5060	UDP/TCP	No	VoIP signaling
24	SIP-TLS	Application	5061	TCP	Yes	Secure VoIP
25	LDAP	Application	389	TCP/UDP	No	Directory services
26	LDAPS	Application	636	TCP	Yes	Secure directory
27	SMB	Application	445	TCP	No	File sharing
28	CIFS	Application	445	TCP	No	Windows file sharing
29	NFS	Application	2049	TCP/UDP	No	File sharing
30	TFTP	Application	69	UDP	No	Lightweight file transfer
31	MQTT	Application	1883	TCP	No	IoT messaging
32	MQTT-TLS	Application	8883	TCP	Yes	Secure IoT
33	AMQP	Application	5672	TCP	No	Message queuing
34	AMQPS	Application	5671	TCP	Yes	Secure messaging

35	CoAP	Application	5683	UDP	No	IoT
36	CoAPS	Application	5684	UDP	Yes	Secure IoT
37	XMPP	Application	5222	TCP	Yes	Messaging
38	IRC	Application	6667	TCP	No	Chat
39	NNTP	Application	119	TCP	No	News groups
40	NNTP-S	Application	563	TCP	Yes	Secure news
41	RDP	Application	3389	TCP/UDP	Yes	Remote desktop
42	VNC	Application	5900	TCP	Optional	Remote control
43	WebSocket	Application	80/443	TCP	Optional	Real-time apps
44	gRPC	Application	443	TCP	Yes	API communication
45	QUIC	Transport	443	UDP	Yes	Modern web transport
46	HTTP/2	Application	443	TCP	Yes	Faster web

47	HTTP/3	Application	443	UDP	Yes	QUIC-based web
48	SOAP	Application	80/443	TCP	Yes	Web services
49	REST	Application	80/443	TCP	Yes	APIs
50	GraphQL	Application	443	TCP	Yes	API queries
51	IP	Network	N/A	N/A	No	Packet routing
52	IPv4	Network	N/A	N/A	No	Internet addressing
53	IPv6	Network	N/A	N/A	Optional	Modern addressing
54	ICMP	Network	N/A	N/A	No	Diagnostics
55	ICMPv6	Network	N/A	N/A	No	IPv6 diagnostics
56	ARP	Data	N/A	N/A	No	MAC resolution
57	RARP	Data	N/A	N/A	No	Reverse ARP
58	MPLS	Network	N/A	N/A	No	Traffic engineering

59	GRE	Network	N/A	N/A	No	Tunneling
60	IPSec	Network	500/4500	UDP	Yes	VPN security
61	L2TP	Data	1701	UDP	No	VPN
62	PPTP	Data	1723	TCP	Weak	VPN
63	OpenVPN	Application	1194	UDP/TCP	Yes	VPN
64	WireGuard	Network	51820	UDP	Yes	VPN
65	BGP	Network	179	TCP	Optional	Routing
66	OSPF	Network	N/A	IP	No	Routing
67	RIP	Network	520	UDP	No	Routing
68	EIGRP	Network	N/A	IP	No	Routing
69	IS-IS	Network	N/A	N/A	No	Routing
70	STP	Data	N/A	N/A	No	Loop prevention

71	RSTP	Data	N/A	N/A	No	Fast STP
72	VLAN (802.1Q)	Data	N/A	N/A	No	Network segmentation
73	PPP	Data	N/A	N/A	No	Point-to-point
74	PPPoE	Data	N/A	N/A	No	DSL connections
75	Ethernet	Data	N/A	N/A	No	LAN communication
76	Wi-Fi (802.11)	Data	N/A	N/A	Yes	Wireless LAN
77	Bluetooth	Data	N/A	N/A	Yes	Short-range comm
78	Zigbee	Data	N/A	N/A	Yes	IoT
79	Z-Wave	Data	N/A	N/A	Yes	IoT
80	CAN	Data	N/A	N/A	No	Automotive
81	Modbus	Application	502	TCP	No	Industrial
82	DNP3	Application	20000	TCP/UDP	Optional	SCADA

83	BACnet	Application	47808	UDP	No	Building automation
84	OPC UA	Application	4840	TCP	Yes	Industrial
85	Kerberos	Application	88	TCP/UDP	Yes	Authentication
86	RADIUS	Application	1812	UDP	Yes	AAA
87	TACACS+	Application	49	TCP	Yes	AAA
88	Diameter	Application	3868	TCP	Yes	AAA
89	SAML	Application	443	TCP	Yes	Authentication
90	OAuth	Application	443	TCP	Yes	Authorization
91	OpenID	Application	443	TCP	Yes	Identity
92	NetBIOS	Application	137-139	TCP/UDP	No	Windows networking
93	mDNS	Application	5353	UDP	No	Local discovery
94	LLMNR	Application	5355	UDP	No	Name resolution

95	SSDP	Application	1900	UDP	No	Device discovery
96	UPnP	Application	1900	UDP	No	Auto config
97	Syslog	Application	514	UDP	No	Logging
98	Syslog-TLS	Application	6514	TCP	Yes	Secure logging
99	NetFlow	Application	2055	UDP	No	Traffic monitoring
100	sFlow	Application	6343	UDP	No	Traffic sampling